Workload-State Verification and Security-Aware Settlement for Spatially Coupled Data Centers

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***Abstract*—Artificial-intelligence data centers can provide demand response, but a meter reduction alone establishes neither workload commitment nor positive grid value. This paper develops a declaration-indexed verification and settlement framework that connects risk budgets, executable jobs, and network value. An exact convex accuracy constraint and an active tail-risk budget define a validation-fitted contract. A residual certificate bounds risk and accuracy changes during execution; an integer formulation jointly enforces continuous job runtimes, site capacity, scenario risk, and exact lattice-based error. Signed nodal valuation includes spatial rebound, while a closed-meter intersection limits payable credit. In a retrospective 54-day comparison, the active-tail formulation reduces mean false credit from 0.565 to 0.540 MWh/day relative to its matched total-only control; the tail difference remains statistically unresolved. Twenty-four coupled runtime cases attain zero optimization gap, and phase-I solves identify the accuracy attainable under binding capacity. A common-domain AC audit retains all 10,332 cells, including 690 numerical unknowns, rather than changing the comparison domain. These results establish explicit contract-admission conditions and reproducible security accounting. Public traces support modeled and observational audits; prospective utility-event evidence is still required for deployment validation.**

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. Introduction

Data centers can shift computing work across time and regions, creating a software-defined source of demand response (DR). Yet a decrease at one meter may coincide with deferred consumption or a load increase at another node. Positive-only local accounting can therefore reward a redistribution with little or negative grid value. Public traces expose multi-week bursty arrivals and scheduler jobs, but measure GPU energy at a different boundary from whole-facility demand [1], [2]. A useful verification protocol must connect these boundaries without equating them.

Historical baselines remain common in market practice [3]; strategic consumption can manipulate their rewards [4]. A meter estimator alone cannot establish that submitted jobs could generate its predicted profile. Workload scheduling and computation–power models already establish temporal deferral, geographical balancing, and grid coordination [5], [6]. Han et al. incorporate conditional value at risk (CVaR) into cross-regional dispatchable-capacity assessment [7]. Here the risk object is instead verification loss: total and daily-tail false credit, subject to an accuracy neighborhood on the same declaration-based profiles.

Network-aware remuneration is also established. Chen et al. combine IDC-aware optimal power flow, locational prices, net-benefit activation, and benefit redistribution [8]. Virtual links remunerate space–time transfers using nodal price differences [9], [10]; receding-horizon control retains multi-site state and future arrivals [11]. Consequently, spatial scheduling, CVaR, and network compensation alone are not our novelty claim. We test a declaration-indexed interface connecting these layers, including where a guarantee does and does not survive conversion between aggregate verification, discrete execution, and settlement.

The study makes three contributions:

1) A typed workload-to-network interface preserves declared job identity, energy, release/deadline feasibility, and regional mapping. Its explicit information gate separates deployable commitments from complete-ledger ex-post comparators; a closed meter is required for payment.
2) A joint total-and-tail risk contract is combined with a contiguous-runtime scheduler and a residual-based composition certificate. This distinguishes optimality for a declared scheduling objective from preservation of the statistical target, risk budgets, or accuracy guarantee. Matched-gate and capacity-stress tests examine the assumptions needed for deployment.
3) Signed nodal and N–1 value are evaluated on the resulting workload profiles, with a finite-scenario value cap and separate meter-backed capacity payment. Common-denominator security checks preserve numerical failures instead of screening them out after observing method outcomes.

The ambition is an auditable end-to-end DR contract, not a collection of unrelated optimization certificates. Standard LP, convex-risk, dispatch, and allocation primitives retain their established meanings [12], [13], [14]. Whether their assumptions hold simultaneously is an experimental question. The public data lack utility-event labels and geographic locations; spatial results are conditional benchmark evidence. Scheduler provenance is trusted: a digest detects changes, not semantically false declarations.

Section II defines the contract objects; Section III gives the verification and composition method; Sections IV and V describe the locked tests and findings. Section VI concludes. The repository supplement retains the detailed audit index.

II. System Model and Problem Formulation

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol h is reserved for a calendar day in statistical panels. The submit-time declaration ledger supplies arrival energy a_{skt} for source

s, class k, and release interval t. The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ. Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^{t} \sum_{d \in \mathcal{D}} x_{skd\tau}. \tag{1}$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^{t} a_{sk\tau}$ and deadline D_k, release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \tag{2}$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \tag{3}$$

The state recursion is $y_{skt} - y_{sk,t-1} = \sum_d x_{skdt}$ with $y_{sk,-1} = 0$, and terminal service equals the declared arrivals due in the window; the release and deadline inequalities are (2) and (3), and the site-capacity constraint is $\sum_{s,k} x_{skdt} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\mathrm{fix}} + \Delta t^{-1} \sum_{s,k} x_{skdt}. \tag{4}$$

For rolling window end T^+ and final event-day release T^0, the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk,\max\{T^0, T^+ - D_k\}}, \tag{5}$$

Equation (5) completes every job due inside the window and every event-day arrival while retaining later real arrivals. The window starts $D_{\max} + 96$ intervals before and ends $D_{\max}$ intervals after the event; omitted earlier work has expired. Equations (2)–(4) map an aggregate preemptive envelope to the facility meter. This is valid for checkpointable batch work; Experiment 14 separately audits contiguous execution intervals, runtime, GPU count, and native power. The indexed declaration ledger contains each job's identifier, release, requested GPUs, declared runtime, class, and region. Its decision information set contains only records available at commitment, topology, tariffs, and configuration. Start/end times and measured energy enter a separate execution ledger after the contract is fixed. Canonical scheduler tuples use the Secure Hash Algorithm 256 (SHA-256) [15]; their integrity is distinct from semantic truthfulness. Complete-ledger profiles use future submissions and are ex-post information comparators. Only the gated protocol can represent a pre-event commitment.

B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, p denotes regional power in MW and time sums use Δt (MWh). The declaration-causal estimate is $\widetilde{p}$; p^0 is the declaration-only strategic reference used for mechanism isolation, $p^{1,\mathrm{sim}}$ is the independently solved event-response schedule, and p^{obs} is the Massachusetts Institute of Technology (MIT) Data Center GPU Manager (DCGM) execution trace. The observed trace is opened only after a profile is frozen; no utility event label exists, so observational replay cannot identify the no-event counterfactual. The mechanism-isolation response $r = p^0 - p^{1,\mathrm{sim}}$ is signed and retains rebound. For $z \in \{\mathrm{obs}, \mathrm{sim}\}$, submitted and model-reference credit are $\widehat{r}^{(z)} = [\widetilde{p} - p^{(z)}]_+$ and $r^{(z)} = [p^0 - p^{(z)}]_+$; false credit is

$$F^{(z)} = \Delta t \sum_{d,t \in \mathcal{E}} [\widehat{r}_{dt}^{(z)} - r_{dt}^{(z)}]_+, \qquad z \in \{\mathrm{obs}, \mathrm{sim}\}. \tag{6}$$

The simulated source defines the validation epigraph; the observed source is recomputed only in locked replay. False credit is computed from (6) in absolute MW-slots, avoiding a ratio with a zero-credit denominator. Before the event the operator freezes a contract baseline p^{con}. After meter closure, define the submitted gross reduction, the contract-cap reduction, and the payable reduction by

$$\begin{aligned} q_{dt}^{\mathrm{gross}} &= [\widetilde{p}_{dt} - p_{dt}^{\mathrm{obs}}]_+, \\ q_{dt}^{\mathrm{cap}} &= [p_{dt}^{\mathrm{con}} - p_{dt}^{\mathrm{obs}}]_+, \\ q_{dt}^{\mathrm{pay}} &= \min\{q_{dt}^{\mathrm{gross}}, q_{dt}^{\mathrm{cap}}\}. \end{aligned} \tag{7}$$

The payable response energy is $Q_{\mathrm{MWh}}^{\mathrm{pay}} = \Delta t \sum_{d,t \in \mathcal{E}} q_{dt}^{\mathrm{pay}}$, and its capacity cash component is $G^{\mathrm{cap}} = r^{\mathrm{DR}} Q_{\mathrm{MWh}}^{\mathrm{pay}}$. Equation (7) uses only the submitted baseline, frozen contract, and closed meter. Here $p^{\mathrm{con}} = p^{\mathrm{safe}}$ for the payment-band panel and is the no-tariff committed-ledger profile for the decision-time panel; p^0 remains an offline diagnostic. On locked event slots, $\mathrm{nRMSE} = \sqrt{N_E^{-1} \sum_n (\widetilde{p}_n - p_n^0)^2} / \max\{N_E^{-1} \sum_n p_n^0, \epsilon\}$. With $M = \sum_n \min\{\widehat{r}_n, r_n\}$, precision, recall, and F_1 are $M / \max\{\sum_n \widehat{r}_n, \epsilon\}$, $M / \max\{\sum_n r_n, \epsilon\}$, and $2PR / \max\{P + R, \epsilon\}$ over the N_E region–slot samples. The metrics are mean absolute error (MAE), root-mean-square error (RMSE), and its normalized form nRMSE; the F1 score (harmonic mean of precision and recall) uses the credit overlap above. The original risk fit is a quadratic program (QP); execution and the exact-error diagnostic use the mixed-integer models below.

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q, and DR price π^{DR}. If reference day j adds event-window service δR_j, the expected payment change is

$$\Delta\Pi = \frac{q\pi^{\mathrm{DR}}}{H} \sum_{j=1}^{H} \delta R_j - \sum_{j=1}^{H} \Delta C_j. \tag{8}$$

Let c_j' be the right derivative of minimum operating cost with respect to the event-window service constraint on day j. Linear-program sensitivity [12] gives

$$\frac{q\pi^{\mathrm{DR}}}{H} > \min_j c_j' \quad \Longleftrightarrow \quad \exists \text{ a strictly profitable local deviation.} \tag{9}$$

The right marginal cost in (9), rather than an arbitrary optimal dual, handles degeneracy. Independent finite differences verify it in Experiment 1. Equations (8)–(9) specialize [4].

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p_t^{\mathrm{native}} + M p_t^{\mathrm{dc}}$ the nodal demand vector. For generator-to-bus map C_g, generator bounds, and power-transfer distribution factor (PTDF) H with line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \ & \sum_i C_i(g_i) \\ \text{s.t. } & \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g} \leq g \leq \bar{g}, \\ & -\bar{f} \leq H(C_g g - p) \leq \bar{f}. \end{aligned} \tag{10}$$

The incidence mapping aggregates demand at its connection bus; (10) is the standard lossless DC representation [16], [17]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \tag{11}$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. Proposed Methodology

A. Model-based framework and matched-information pre-estimation

The state, power, and network equations define the feasible state before a payment target is formed (Fig. 1). Meter history and submitted declarations enter the estimator in parallel; execution remains scoring-only. Lagged meter/calendar baselines and declaration-view ledger comparators include ridge, gradient boosting, and extremely randomized trees [18], [19], [20]. Synthetic control and nonnegative donor weighting are evaluated as a separate history-based class [21]. The quantile output receives the same exact projection, with its penalty selected by contiguous folds.

B. Sparse cumulative workload-state verifier

For statistical estimate $\widetilde{p}$ and each predeclared penalty ρ_ℓ, the first-stage program solves

$$\min_{x,y,z} \ \mathcal{C}_{\mathrm{w}}(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \geq \pm(p_{dt}(x) - \widetilde{p}_{dt}), \tag{12}$$

subject to (1)–(4), with workload cost $\mathcal{C}_{\mathrm{w}}$. The epigraph (12) [12] changes the objective, not feasibility. Six globally solved metadata schedules $p^{(\ell)}$ share identical arrivals and constraints. A seventh causal Pareto projection is selected on validation from the high-5-of-10 declaration target by a lexicographic rule over nRMSE, false credit, under-credit, and $1-F_1$. A complete-ledger feasible-quantile projection is retained as an external post-commitment comparator and is used only in the separate payment hull. For a hull profile, $\bar{p} = \sum_{\ell=1}^{L} \alpha_\ell p^{(\ell)}$, $\alpha \geq 0$, $\mathbf{1}^\top \alpha = 1$, remains feasible because the common scheduling set is convex.

The convex fit minimizes validation error plus total and daily-tail false-credit exposure, measured by an empirical top-m tail mean (the CVaR epigraph at effective level $1 - m/H$) [13]. Let $\ell^\star$ minimize contiguous-fold nRMSE and let B_h be the absolute daily false-credit exposure of the validation-selected causal Pareto anchor on validation day h. This anchor differs from the external single-projection comparator. For a candidate simplex profile define $F_h(\alpha) = \sum_{n \in h} f_n$, where f_n is the positive-part false credit at sample n. Exposure is in MW-slots, without division by true credit. With $m = \lceil (1-\gamma)H \rceil$ and reserve $\beta \in (0, 1]$, solve

$$\begin{aligned} \min_{\alpha, f, \nu, u} \quad & \frac{\left\|\bar{p} - p^{0,\mathrm{decl}}\right\|_2^2}{N s_p^2} + \lambda \|\alpha\|_2^2 + \omega_T \frac{\sum_{h=1}^{H} F_h}{S_T} \\ & + \omega_C \frac{\nu + m^{-1} \sum_{h=1}^{H} u_h}{S_C}, \end{aligned} \tag{13}$$

In (13), $s_p^2 = \max\{N^{-1}\|p^{0,\mathrm{decl}}\|_2^2, \epsilon\}$, $S_T = \max\{\sum_h B_h, \epsilon\}$, and $S_C = \max\{m^{-1} \sum_{h \in \mathrm{tail}_\gamma} B_h, \epsilon\}$. Single-axis ablations remove one term and its constraint; the locked joint fit uses $\omega_T = 1.00$, $\omega_C = 0.50$, $\gamma = 0.75$, and $\lambda = 10^{-8}$. Both preferences and the total/tail budgets (14)–(15) are frozen before locked scoring:

$$\sum_{h=1}^{H} F_h(\alpha) \leq \beta \sum_{h=1}^{H} B_h, \tag{14}$$

$$\mathrm{Tail}_m(F_h(\alpha)) \leq \beta \mathrm{Tail}_m(B_h). \tag{15}$$

The positive part is represented without differentiating through a sort. For each event sample n and validation day h, introduce $f_n \geq 0$, a CVaR threshold $\nu \geq 0$, and tail slacks $u_h \geq 0$:

$$f_n \geq p_n^{\alpha} - p_n^{(\mathrm{sim})} - r_n^{(\mathrm{sim})}, \tag{16}$$

$$F_h(\alpha) = \sum_{n \in h} f_n, \tag{17}$$

$$F_h(\alpha) - \nu - u_h \leq 0, \tag{18}$$

$$\nu + \frac{1}{m} \sum_{h} u_h \leq \beta \, \mathrm{Tail}_m(B_h). \tag{19}$$

Equation (19) uses the same top-m convention as scoring. Here $r_n^{(\mathrm{sim})} = [p_n^0 - p_n^{(\mathrm{sim})}]_+$, so its false-credit row equals the nested-positive-part definition. To preserve the point-estimate metric, let $q_\ell = N^{-1/2}\|p^{(\ell)} - p^{0,\mathrm{decl}}\|_2$ and $q_\star$ denote the RMS error of the validation-selected causal Pareto reference. The simplex additionally satisfies the linear sufficient condition

$$\sum_{\ell=1}^{L} \alpha_\ell q_\ell \leq \sqrt{1 + \varepsilon_{\mathrm{acc}}}\, q_\star, \qquad \varepsilon_{\mathrm{acc}} = 0.10. \tag{20}$$

The triangle inequality proves (20) sufficient for 10% validation-MSE non-inferiority. The sparse QP is solved with SciPy trust-constr; Karush–Kuhn–Tucker stationarity and all primal residuals are recomputed. Four contiguous folds select β only when every training contract is feasible. A separate payment projection targets $\bar{p}$ under

$$p_{dt}^{\mathrm{safe}} \leq p_{dt}^{(\ell^\star)}. \tag{21}$$

This event-slot cap (21) never replaces the risk-fitting profile.

a) Binding-tail revision test.: To distinguish a hard risk constraint from a soft objective preference, Experiment 31 keeps the candidate hull and 10% accuracy neighborhood but replaces (20) with the exact convex row $N^{-1}\|\bar{p} - p^0\|_2^2 \leq (1 + \varepsilon_{\mathrm{acc}}) q_\star^2$. Starting from the exact-MSE total-only optimum, the validation total budget is 1.10 times its exposure. A second solve minimizes the top-m tail under that budget. The primary tail cap is the midpoint between this minimum and the total-only tail; quarter- and three-quarter points provide the frozen frontier. Its objective retains only MSE and total exposure, so an active tail constraint has an identifiable role. A separate control removes the old hard risk rows while preserving the old objective. These are retrospective revision experiments motivated by previously inspected locked results, not a new untouched holdout.

b) Two-sided credit band.: To exclude a zero-credit envelope, tie a lower band to the fitted target. Define $p_{dt}^{\mathrm{floor}} = \min\{p_{dt}^{(\ell^\star)}, \bar{p}_{dt}\}$, where $\bar{p}$ is the pre-projection validation fit. The event-slot constraints are

$$p_{dt}^{\mathrm{floor}} - \varepsilon \leq p_{dt}^{\mathrm{safe}} \leq p_{dt}^{(\ell^\star)}, \qquad (d, t) \in \mathcal{D} \times \mathcal{E}, \tag{22}$$

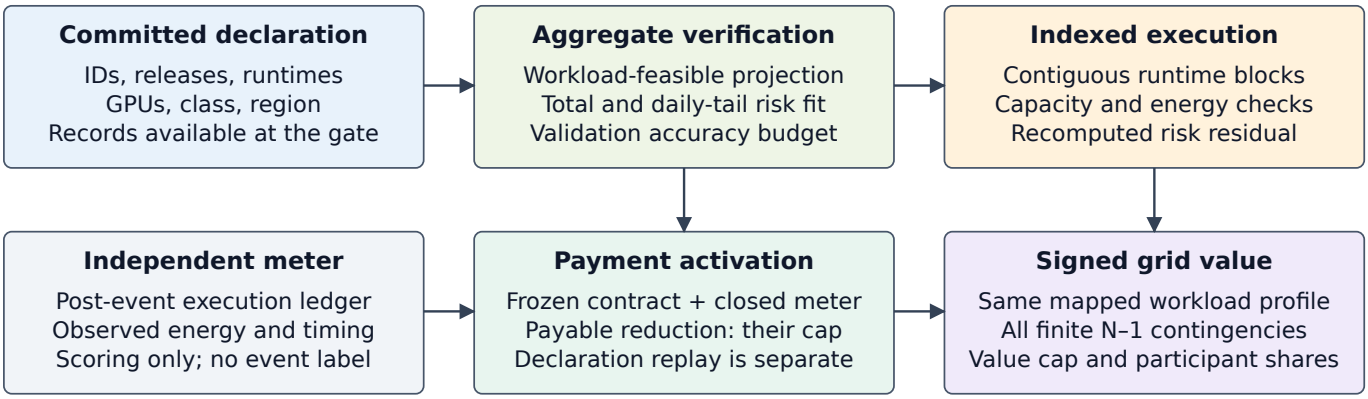


Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N–1 value and exact participant shares.

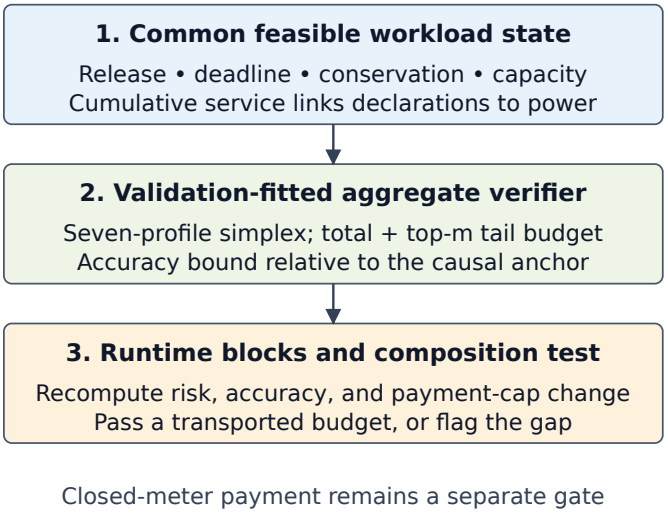


Fig. 2. Verification and execution stages. Aggregate feasibility, validation risk, and runtime feasibility are distinct certificates. The composition test recomputes their compatibility before network valuation or payment.

In (22) we set $\varepsilon = 0$; the upper profile proves nonemptiness. The lower side regularizes under-credit without an unseen-trace guarantee. The separate meter cap (7) controls realized payable credit.

c) Decision-time information boundary.: A gate removes all post-gate arrivals before refitting the matched-information target. The locked protocol uses slot 62, 30 minutes before the first event slot, and a no-tariff LP freezes the contract baseline with a predeclared 3-MWh minimum of participating event-window service. A second LP applies the selected positive DR price to the same masked ledger while carrying unexpired state. Reserved future jobs cannot be credited, and the tariff plan is scored against the closed meter through (7). The independent execution trace is read only in the scoring pass.

An earlier-days-only quantile reserve is evaluated separately for planning; unsubmitted future jobs are never payment eligible. Sparse cumulative states have $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$ nonzeros. Two lexicographic LPs minimize L1 error and then operating cost using HiGHS [22]. Figure 2 summarizes the stages.

C. Runtime-complete execution and composition

Let job j have release r_j, requested GPUs g_j^{req}, declared runtime R_j slots, fixed region s_j, and service endpoint $d_j = r_j + R_j + B$, where B is a precommitted queue allowance. With GPU nameplate $\bar{p}_{\mathrm{GPU}}$, Define the runtime-complete entitlement $E_j^{\mathrm{run}} = R_j g_j^{\mathrm{req}} \bar{p}_{\mathrm{GPU}} \Delta t$. The indexed service variable $u_{j\tau}$ is the energy assigned to job j in interval τ, and the executable witness requires one contiguous fixed-rate block of exactly R_j slots. Let w_{batch} be the predeclared waiting-cost coefficient per MWh-slot, and r^{DR} the event service tariff. Define $c_j = g_j^{\mathrm{req}} \bar{p}_{\mathrm{GPU}} \Delta t$ and $\Theta_j = \{r_j, \dots, d_j - R_j\}$. For each admissible start θ, a binary $z_{j\theta}$ selects the block and $u_\tau^{j\theta} = c_j \mathbf{1}\{\theta \le \tau < \theta + R_j\}$. The exact start-time mixed-integer linear program (MILP) is

$$\begin{aligned} &\sum_{\theta \in \Theta_j} z_{j\theta} = 1, \qquad z_{j\theta} \in \{0,1\}, \\ &u_{j\tau} = \sum_{\theta \in \Theta_j} u_\tau^{j\theta} z_{j\theta}, \qquad \sum_{j : s_j = d} u_{j\tau} \le \bar{P}_d \Delta t, \\ &\min_{z} \sum_{j, \theta \in \Theta_j} z_{j\theta} \sum_{\tau} \left[w_{\mathrm{batch}} (\tau - r_j) + r^{\mathrm{DR}} \mathbf{1}\{\tau \in \mathcal{E}\} \right] u_\tau^{j\theta}. \end{aligned} \tag{23}$$

Every selected block in (23) enforces runtime completion, contiguity, and the GPU nameplate. Independent per-job enumeration solves the displayed MILP exactly only if the combined minimizers satisfy every regional capacity row. Binding capacity requires the coupled MILP; separability is not presumed. Waiting cost is charged at every service interval, not only the start. The training-calibrated entitlement $\widehat{E}_j$ is carried on the same blocks only as a diagnostic; it is not measured consumption and does not replace the runtime/nameplate commitment. The risk target is linked through a fixed regional slot price. Because the statistical and nameplate ledgers have different scales, define the locked window set $\mathcal{W}_{\mathrm{lock}} = \bigcup_{h \in \mathcal{H}_{\mathrm{lock}}} \{96h, \dots, 96h + 95\}$. The declaration energy active in this set is

$$E_{\mathrm{act}}(\mathcal{W}_{\mathrm{lock}}) = \sum_{j \in \mathcal{J}_{\mathrm{sub}}} E_j^{\mathrm{dec}} \frac{|[r_j, r_j + R_j) \cap \mathcal{W}_{\mathrm{lock}}|}{R_j}, \tag{24}$$

In (24), E_j^{dec} is either the nameplate upper entitlement or the calibrated central entitlement. This overlap integral includes carry-in jobs and uses only submit-time fields. Let $\widetilde{p}^{\mathrm{risk}}_{dht} = P^{\mathrm{fix}}_d + \gamma_+[\bar{p}^{\mathrm{risk}}_{dht} - P^{\mathrm{fix}}_d]_+$, with $\gamma_+ = E_{\mathrm{act}}(\mathcal{W}_{\mathrm{lock}})/\sum_{d,h,t}[\bar{p}^{\mathrm{risk}}_{dht} - P^{\mathrm{fix}}_d]_+\Delta t$. For locked day h, region d, and interval t, define the fixed normalized slot-price heuristic $\pi_{dht} = r^{\mathrm{DR}}[\widetilde{p}^{\mathrm{risk}}_{dht} - P^{\mathrm{fix}}_d]_+/\widetilde{\Pi}_{\max}$, where $\widetilde{\Pi}_{\max} = \max_{d,h,t}[\widetilde{p}^{\mathrm{risk}}_{dht} - P^{\mathrm{fix}}_d]_+$. Re-enumerating every declaration-feasible block with the waiting, event-tariff, and π_{dht} terms produces the exact minimizer of this finite linear-price objective. The optimality statement concerns the declaration-feasible policy; equality to the aggregate target is a separate primal residual. The bridge therefore records, for every locked day,

$$\begin{aligned} e^{\mathrm{br}}_{dht} &= p^{\mathrm{exec}}_{dht} - \widetilde{p}^{\mathrm{risk}}_{dht}, \\ \mathrm{nRMSE}_{\mathrm{flex}} &= \frac{\|e^{\mathrm{br}}\|_2}{\max\{\|\widetilde{p}^{\mathrm{risk}} - P^{\mathrm{fix}}\|_2, \epsilon\}}. \end{aligned} \tag{25}$$

Equation (25) defines the executable-profile residual and its flexible relative-L2 measure, alongside the total-load normalization. The positive normalization cancels from the slot price, so active-window scaling changes target magnitude but not the selected starts. A high target load receives a high positive cost; this is the objective of the archived slot-price comparator; it is not a derived target-tracking dual price. The resulting risk and accuracy must be recertified after execution. For the job-to-network certificate, the indexed witness is mapped without a second optimization. To avoid overloading the deadline endpoint d_j, let $\mathcal{J}_{sk\delta}$ be the ledger partition whose source, class, and declared destination labels are s, k, δ; Experiment 27 uses the native assignment $\delta = s_j$ and therefore has no outcome-dependent migration:

$$x^{\mathrm{job}}_{sk\delta\tau} = \sum_{j \in \mathcal{J}_{sk\delta}:\, r_j \le \tau < d_j} u_{j\tau}, \tag{26}$$

$$p^{\mathrm{job}}_{\delta\tau} = P^{\mathrm{fix}}_\delta + \Delta t^{-1} X^{\mathrm{job}}_{\delta\tau}, \qquad X^{\mathrm{job}}_{\delta\tau} = \sum_{s,k} x^{\mathrm{job}}_{sk\delta\tau}. \tag{27}$$

Equation (27) is reconstructed from the indexed blocks before valuation; no independently refitted aggregate receives the job witness's certificate. Aggregate migration remains a separate control. Deferral and migration costs follow [5], [23], [11].

a) Residual-to-risk composition.: Fix a baseline b, executable profile x, model reference r, and meter m on the same samples and scale. The nested positive parts in (6) simplify to $F_h(b) = \Delta t \sum_{n\in h}[b_n - \max\{m_n, r_n\}]_+$. Let $\delta_h = \Delta t \sum_{n\in h} |x_n - b_n|$ and let T_m average the largest m daily values. Then

$$F_h(x) \le F_h(b) + \delta_h, \tag{28}$$

$$\sum_h F_h(x) \le \sum_h F_h(b) + \sum_h \delta_h,$$

$$T_m(F(x)) \le T_m(F(b)) + T_m(\delta), \tag{29}$$

$$\mathrm{RMSE}(x, r) \le \mathrm{RMSE}(b, r) + \mathrm{RMSE}(x - b). \tag{30}$$

The positive part is 1-Lipschitz, proving (28). Summation gives the total bound; monotonicity and subadditivity of the top-m mean give (29); (30) is the triangle inequality. Thus residual budgets can certify transport, while an exact recomputation can pass when the sufficient bound is conservative. Identity and energy conservation alone cannot imply these inequalities meet a specified risk budget.

For payable credit $q = \min\{[x - m]_+, [b - m]_+\}$, its false credit is bounded pointwise by that of b. Therefore the intersection preserves total and tail risk whenever b passes on the same reference, meter, and sample set. It does not preserve executable-profile accuracy or identify a physical no-event counterfactual. Experiment 31 checks both the uncapped execution certificate and the separate capped-credit certificate without conflating their outcomes.

b) Direct risk-constrained runtime optimization.: The composition inequalities permit failure; a stronger alternative puts the risk rows inside the full-runtime MILP. Experiment 32 fixes declaration-only reference and response scenarios, adds total and top-m risk rows, and tests an exact MSE budget on the same jobs. For a common GPU power quantum q, normalized integer load z and normalized target ζ admit the exact squared-error epigraph

$$v \ge (2k - 2\zeta)z + \zeta^2 - k^2, \quad k = 0, \dots, \bar{z}. \tag{31}$$

For any feasible integer z, the row $k = z$ attains $(z - \zeta)^2$, while every other row is below it. Thus (31) with an MSE budget or minimum-MSE objective enforces the exact error; the physical MSE is $q^2 \sum v/N$ over N event samples. The diagnostic retains infeasible contracts and minimizes achievable MSE under unchanged risk/capacity rows. It is a finite known-scenario certificate, not foresight of an unknown future meter.

D. Signed nodal and N–1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^\top(p^0 - p^1)$, while (11) is the realized signed system-value benchmark. Convexity [12] gives

$$0 \le (\lambda^0)^\top(p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^\top(p^1 - p^0). \tag{32}$$

Thus the linear rule upper-bounds exact value in the polyhedral market model; equality requires a common supporting subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from anticipatory shift through recovery. Its space–time remuneration is

$$\Pi^{\mathrm{ST}} = \Delta t \sum_{t\in\mathcal{W}} \sum_d \lambda^0_{dt}(p^0_{dt} - p^1_{dt}). \tag{33}$$

This is the nodal virtual-link structure of [9], [10]; summing site payments reproduces (33).

Space–time energy remuneration is one component of the contracted DR product. Let $\mathcal{D}^{\mathrm{part}}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and r^{DR} the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\mathrm{cap}} = \Delta t \sum_{d\in\mathcal{D}^{\mathrm{part}}} \sum_{t\in\mathcal{E}} (p^{\emptyset}_{dt} - p^1_{dt}), \tag{34}$$

$$G = r^{\mathrm{DR}} Q^{\mathrm{cap}} + \Pi^{\mathrm{ST}}, \tag{35}$$

where $p^{\emptyset}$ is the minimum-cost no-event schedule on the same window. Equation (34) defines the bilateral capacity product and need not equal trace-anchored p^0; Π^{ST} debits spatial transfer and recovery. This signed service model follows demand-response mechanism design [8], [24].

For operator value (35), define opportunity cost $C_{\mathrm{opp}} = \mathcal{C}_{\mathrm{w}}(x^1) - \mathcal{C}_{\mathrm{w}}(x^{\emptyset})$ and surplus $S = G - C_{\mathrm{opp}}$. A symmetric Nash contract [25] activates iff $S \geq 0$, transfers $C_{\mathrm{opp}} + S/2$, and gives each party utility $S/2$. Nonactivation is the zero-utility outside option. This bilateral transfer is separate from nodal remuneration and exact dispatch value. For the full-day ledger, the interval system amount is $s_t = r^{\mathrm{DR}}\Delta t \mathbf{1}\{t \in \mathcal{E}\}\sum_d q_{dt}^{\mathrm{pay}} + \Delta V_t$, where q^{pay} is in MW, Δt is in hours, and $\Delta V_t = \Delta t[V(p_t^0) - V(p_t^1)]$ is in USD. The economic ledger $S^{\mathrm{signed}} = \sum_{t\in\mathcal{W}} s_t$ retains debits; the separate cash presentation is $S^{\mathrm{cash}} = \sum_{t\in\mathcal{W}}[s_t]_+$. The floor is applied to each system interval, not to independently allocated regional values. Complete-cycle conservation gives $\sum_{d,t\in\mathcal{W}}(p_{dt}^0 - p_{dt}^1)\Delta t = 0$ within solver tolerance.

Before payment comparison, the submitted, frozen-contract, and closed-meter reductions are intersected as follows:

$$q_{dt}^{\mathrm{pay}} = \min\{q_{dt}^{\mathrm{obs}}, q_{dt}^{\mathrm{cap}}\}, \qquad G = r^{\mathrm{DR}}\Delta t \sum_{d,t\in\mathcal{E}} q_{dt}^{\mathrm{pay}} + \Pi^{\mathrm{ST}}. \tag{36}$$

Equation (36) is evaluated after meter closure. Experiment 12 audits its budget balance; Experiment 27 instead stores $q_{dt}^{\mathrm{replay}} = \min\{q_{dt}^{\mathrm{dec}}, q_{dt}^{\mathrm{cap}}\}$ to certify feasible network value. Experiment 30 records this role as declaration-only, and the replay's signed/cash totals remain outside payable settlement until q^{obs} is available.

Security-aware settlement augments (10) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ, then

$$-\bar{f}_\ell \leq (H_\ell + L_{\ell k}H_k)(C_g g - p) \leq \bar{f}_\ell, \quad \ell \neq k. \tag{37}$$

Equation (37) is the standard preventive N–1 LODF formulation [26]; every credible row uses native ratings. Islanding rows are excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the DC/N–1 planning scope [27].

E. Finite-scenario N–1 value-component certificate

The pointwise envelope controls energy but not network payment because spatial loads can activate different security constraints. Let $\mathcal{P} = \mathrm{conv}\{p^{(1)}, \ldots, p^{(L)}\}$ be the feasible-profile hull and $p^{\mathrm{cap}} \in \mathcal{P}$ the validation-selected projection. Its penalty is separate from the payment-target penalty; the feasible-quantile projection is an external comparator. Let Ξ_5 be the 1st, 10th, 50th, 90th, and 99th calibration-validation GPU-energy ratios from independent telemetry [2]. For facility profile p_{dt}^{fac}, fixed demand P_d^{fix}, scale s_{dc}, and factor ξ,

$$p_{dt}^{\mathrm{dc},\xi} = s_{\mathrm{dc}}\big(P_d^{\mathrm{fix}} + \xi[p_{dt}^{\mathrm{fac}} - P_d^{\mathrm{fix}}]\big), \qquad p_{dt}^{\mathrm{fac}} \geq P_d^{\mathrm{fix}}. \tag{38}$$

The scale uses the calibration-validation q_{99} flexible peak, keeping the fixed component invariant to ξ; raw endpoints remain activation scenarios. The injection is $Mp_t^{\mathrm{dc},\xi}$, with superscripts α and "cap" applying (38) to p^α and p^{cap}. For each settlement day, the value-certified verifier defines $B_\xi^{\mathrm{cap}} = \Delta t \sum_{t\in\mathcal{E}} V_{N-1}(p_t^{\mathrm{native}} + Mp_t^{\mathrm{dc},\xi,\mathrm{cap}})$ and solves

$$\begin{aligned}
\mathrm{lexmin}_{\alpha,e,\eta,\{g_t^\xi\}} \quad & \left(\sum_{d,t\in\mathcal{E}} e_{dt}, -\eta\right) \\
\text{s.t.} \quad & p^\alpha = \sum_{\ell=1}^{L} \alpha_\ell p^{(\ell)}, \\
& \alpha \geq 0, \quad \mathbf{1}^\top\alpha = 1, \quad 0 \leq \eta \leq 1, \\
& e_{dt} \geq \pm(p_{dt}^\alpha - p_{dt}^{\mathrm{safe}}), \\
& g_t^\xi \text{ satisfies (10) and (37)}, \\
& \qquad \text{with } p_t = p_t^{\mathrm{native}} + Mp_t^{\mathrm{dc},\xi,\alpha}, \\
& \qquad\qquad t \in \mathcal{E}, \quad \xi \in \Xi_5, \\
& \Delta t \sum_{t\in\mathcal{E}}\sum_i C_i(g_{it}^\xi) \\
& \qquad \leq (1-\eta)B_\xi^{\mathrm{cap}}, \quad \xi \in \Xi_5.
\end{aligned} \tag{39}$$

The epigraph in (39) is solved by two global LPs: trajectory error is minimized first, then the common fractional margin η. Feasibility is the N–1 upper bound and p^{cap} proves nonemptiness. Hence, for any $\xi \in \Xi_5$ and realized load $p^{1,\xi}$, the daily dispatch-value component obeys

$$\begin{aligned}
&\sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p_t^{\mathrm{native}} + Mp_t^{\mathrm{dc},\xi,\alpha}) - V_{N-1}(p_t^{1,\xi})] \\
&\leq \sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p_t^{\mathrm{native}} + Mp_t^{\mathrm{dc},\xi,\mathrm{cap}}) - V_{N-1}(p_t^{1,\xi})].
\end{aligned} \tag{40}$$

The realized cost cancels, so this is a samplewise value-component certificate without execution truth. Inequality (40) does not bound the capacity term, linear space–time transfer, or positive cash floor of the entire contract. The hull contains six first-stage projections and feasible quantile. The network contract targets the joint-risk profile but is excluded from its fit.

A separate continuous-segment audit minimizes dispatch value over the segment between the frozen cap and certified profile. A joint LP gives its minimum; convexity places the maximum at an endpoint when the realized counterfactual is fixed. This does not establish a continuous-conversion guarantee when the realized load also changes. The repository supplement gives the interval formulation and its independent-resolution checks.

F. Participant allocation

For participants N, let $v(S)$ be signed dispatch value when coalition S moves from baseline to actual demand. The exact allocation is

$$\phi_i(v) = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!}[v(S\cup\{i\}) - v(S)]. \tag{41}$$

The signed allocation (41) [14] is efficient: $\sum_i \phi_i = v(N)$ if $v(\varnothing) = 0$. All coalitions are evaluated for four sites and eight portfolios; exchangeable site slices reduce larger contracts to count states. This allocation does not change physical feasibility or the meter gate.

G. Scope of the composed certificate

The common ledger preserves provenance, not automatically every guarantee. For incidence $B_{dj} = \mathbf{1}\{s_j = d\}$ and bus map M, the implementation checks job energy $\sum_\tau u_{j\tau} = E_j^{\mathrm{run}}$, regional equality $X_{d\tau} = \sum_j B_{dj} u_{j\tau}$, and nodal mapping $p_{b\tau} = P_{b\tau}^{\mathrm{fix}} + \Delta t^{-1} \sum_d M_{bd} X_{d\tau}$. These equalities ensure that the valued profile is the indexed executable profile. If dispatch value is L_V-Lipschitz on the committed feasible polytope, a mapping residual r^{map} incurs at most $L_V \|r^{\mathrm{map}}\|_1$ value error.

The workload projection and its convex hull inherit only the aggregate feasible set. The accuracy bound holds on validation, and the value cap holds only on its frozen conversion scenarios. Runtime completion, information availability, transported risk, network feasibility, and meter activation are therefore checked as distinct predicates on one profile. Passing a digest or an optimizer residual alone is insufficient for end-to-end activation.

IV. Experimental Design

A. Data and predeclared information sets

BurstGPT supplies 5,188,507 successful requests [1]; MIT Supercloud supplies 75,326 valid scheduler submissions and 71,128 positive-energy executions [2]. The aggregate tensor contains 68,662 executions with valid intervals. Chronological fitting uses the first 40 complete days; validation uses days 45–60 and locked scoring uses days 61–114. Submit-time features define decisions. Measured intervals and DCGM energy are opened only for scoring, with no utility-event label.

Aggregate deferral windows are 0, 4, and 512 slots. Each scenario region has a 6-MW fixed component and 118-MW flexible capacity. For the runtime witness, $\bar{p}_{\mathrm{GPU}} = 0.001$ MW/GPU, $B = 96$ slots, and an unlimited Slurm runtime maps to 128 slots; finite declarations are retained. Runtime and queue allowance are separate: Slurm timelimit is not a submit-to-completion deadline. The runtime/nameplate entitlement, training-calibrated central energy, measured GPU energy, and network-scaled facility demand remain separate quantities. Population and calibration joins are documented in the supplement.

Geography is unobserved. Four regions are assigned by a predeclared feature-stratified round robin; the spatial panel exhausts 24 permutations, two concentration controls, and three penetrations at a 0.75 native-load multiplier. Payment-network scaling uses the calibration-validation flexible q_{99} peak at a declared 4% penetration, with the fixed component held separate. Five measured conversion ratios span 0.583–5.321; neither normalization nor nameplate coverage establishes a measured 100-MW facility. Topologies, ratings, and costs follow the Power Grid Library (PGLib), MATPOWER, and RTS-24 [17], [28], [29].

B. Comparators and experimental questions

The aggregate panel compares history-only estimators, declaration-feature estimators, synthetic control, a single feasible projection, and a validation-selected causal Pareto anchor. The proposed joint-risk verifier combines total and daily-tail preferences; "total-only" and "tail-only" are independently solved ablations. The value-certified verifier is a separate network-value cap, never an additional risk-fitting candidate. Complete-ledger quantile comparators are explicitly ex-post. Structural scheduling translations use matched constraints but are not reproductions of published software [5], [30].

Four contiguous leave-block-out validation folds select projection penalties and risk reserve; these are not rolling-origin folds. The joint QP fixes $\omega_T = 1$, $\omega_C = 0.5$, $\gamma = 0.75$, $\lambda = 10^{-8}$, and the 10% validation-MSE budget. Tail statistics always average exactly the worst $m = \lceil (1-\gamma)H \rceil$ days, including in bootstrap samples with ties. Thus the locked 54-day statistic uses 14 days and effective level $1 - 14/54 = 0.741$, rather than exact empirical CVaR at 0.75.

Three-day moving-block percentile intervals are exploratory and unadjusted; block-length sensitivity is reported. The separate exact block-sign tests use all 2^{18} assignments and Holm-adjusted p-values [31], [32]. Means improving together do not imply simultaneous statistical significance. Model- and source-hash checks bind results to the saved profiles; recomputation using a shared selector is distinguished from an independent algorithmic cross-check.

The experiment index supports four questions: whether risk control improves verification; whether the guarantee survives runtime execution under one information set; whether signed network accounting adds value; and whether security results retain a common denominator. The original witness and bridge (Experiments 27–30), composition test (Experiment 31), and matched-gate capacity test (Experiment 32) retain their distinct roles. The AC fixed-plan panel freezes non-reference active generation after intact dispatch, allowing reference balance, reactive output, and voltage recourse. Its six snapshots are days 61, 88, and 114 at slots 64 and 71, not independent operating days. All numerical backends are single-threaded; revision tests use at most two concurrent CPU workers.

V. Results

A. Verification accuracy and risk trade-offs

The mechanism-isolation experiment confirms the economic motivation. At \$150/MWh and call probability 0.35, strategic scheduling increases the historical event baseline by 8.223 MWh, of which 44.458 of 55.452 credited MWh are unsupported by the model-reference reduction. The marginal threshold (9) agrees with independent strategic optimization in all 56 price/probability cells (Experiment 1).

Table I separates the verification trade-off from deployment. The joint-risk verifier lowers mean false credit from 2.470 to 0.780 MWh/day relative to the single projection, and improves its other three means. Its nRMSE improvement is not significant in the reported Holm-adjusted test ($p = 0.158$; three-day interval $[-0.006, 0.080]$). Block-sum autocorrelation is 0.727 for this contrast, so the supplied block-length sensitivity matters. The causal Pareto anchor is more accurate and has less under-credit; the total-only ablation has better means on all four metrics. The data therefore do not establish that adding the old tail preference improves prediction.

The validation joint MSE is 200.575 MW^2, below its 207.825-MW^2 budget; stationarity and primal residuals are 9.69×10^{-10} and 1.11×10^{-16}. Both old risk budgets have

TABLE I
COMPLETE-LEDGER MECHANISM-ISOLATION RESULTS, 54-DAY MEANS. CREDIT COLUMNS ARE MWH/DAY; NRMSE AND F_1 ARE DIMENSIONLESS.

Method	nRMSE	False credit	Under-credit	F_1
High-5-of-10	1.586	41.220	11.169	0.572
History ridge	1.038	21.906	25.226	0.545
History boosting	1.011	20.977	26.401	0.534
History extra trees	1.090	26.629	18.176	0.597
Declaration boosting	0.774	9.183	21.528	0.640
Ex-post decl. boosting	0.750	9.171	20.817	0.643
Ex-post quantile	1.337	0.591	54.391	0.237
Synthetic control	0.782	9.394	21.678	0.676
Feasible quantile	1.344	0.098	52.647	0.225
Single projection	0.943	2.470	26.812	0.463
Causal Pareto anchor	0.691	1.895	17.024	0.699
Total-only	0.892	0.646	26.372	0.549
Tail-only	0.905	0.817	26.541	0.533
Original joint	0.902	0.780	26.506	0.536
Exact-MSE total-only	0.949	0.565	28.481	0.506
Active tail	0.945	0.540	28.384	0.511

Complete-ledger rows are ex-post information comparators. High-5-of-10 is not workload-feasible. The causal anchor and single projection are distinct validation targets. The final two rows use the exact-MSE neighborhood.

positive slack. Removing only those rows with the objective unchanged leaves essentially the same solution, confirming that their presence does not explain the old fit.

The binding-tail revision instead makes the tail row active (slack 4.63×10^{-8}, dual 0.0537). Against its matched exact-MSE total-only control, it reduces locked mean false credit from 0.565 to 0.540 MWh/day and top-14-day tail from 1.944 to 1.861 MWh. Its nRMSE is 0.945 versus 0.949, and F_1 is 0.511 versus 0.506. The paired tail difference has an unadjusted 95% block-bootstrap interval $[-0.182, 0.006]$ MWh, including zero. Compared with the original joint fit, this variant sacrifices accuracy and recall for lower false credit. This is a measured frontier, not a confirmatory superiority result (Experiment 31; Fig. 3, left).

Independent observed-meter replay is less favorable: event-window MAE is 6.880 MW for the joint-risk verifier, 7.038 MW for the single projection, and 1.909 MW for feasible quantile (Experiment 20). Meter independence prevents scoring leakage, but supplies no identified no-event counterfactual.

B. Does the certificate survive execution?

The full runtime witness covers 75,326 declarations and 7,306,622 admissible starts. Saved-policy recomputation has zero start mismatches and an objective residual of 2.27×10^{-13} USD; it reuses the original selector and is not an implementation-independent optimality test. Capacity utilization peaks at only 0.444%, so the scalable exact result concerns a nonbinding regime. The original bridge's total-profile nRMSE is 0.030, but its flexible relative L2 error is 1.581 and maximum discrepancy is 3.220 MW. The fixed 6-MW component masks this mismatch in the total-load normalization.

The scale-aligned composition test makes the failed link explicit (Experiment 31). Under the same archived nameplate affine scale, the target's total false credit is 1.610 MWh and its tail is 0.100 MWh. Execution raises these to 22.101 and 1.149 MWh, exceeding carried audit budgets of 4.168 and 0.235 MWh. Executed MSE is 0.857 MW^2 against a 0.304-MW^2 budget. All three uncapped execution tests fail, despite exact job/network provenance. The total budget scales the validation daily allowance to 54 days; it is an audit threshold, not a generalization theorem.

Intersecting executable and target credit reduces total false credit to 0.804 MWh and tail to 0.055 MWh, passing both risk limits. It also removes 22.981 of 32.829 MWh of uncapped credit. This proves the intended distinction: the payment-risk cap can work while the executable-profile certificate fails. This panel uses a model-generated event meter and a locked-block scale; it remains an ex-post mechanism audit, not deployable meter evidence.

The matched-gate panel uses slot 62, identical eligible arrivals and pre-gate aggregate state, all four sites, a common modeled reference/response, and the same 3-MWh service floor (Experiment 32). Its 13 methods retain all 70 validation/locked days. Joint, total-only, tail-only, and unconstrained fits collapse to the selected causal projection: locked nRMSE 0.062, false credit 0.631 MWh/day, and recall/F_1 0.790. The seven-candidate event hull has rank two and only three distinct profiles, explaining the absence of an incremental risk-module advantage. All methods' payable modeled credit is 3 MWh/day with zero modeled shortfall because of the binding floor and common benchmark state; this is not measured delivery reliability.

The full-runtime capacity scan retains 64-job gate cohorts on days 61, 88, and 114, with 32 pending and 32 already-running jobs each. Hypothetical capacity changes; job power, runtime, deadlines, and all-slot constraints do not. All 24 cost-only/direct-risk cells solve with zero gap, at achieved peak utilizations of 19–99% (Fig. 3, right). Risk rows change one high-load cell's false credit from 0.00025 MWh to zero. The small cohort therefore demonstrates coupled feasibility without repairing the 75,326-job bridge or establishing full-scale congestion performance.

A sequential exact-match diagnostic imposes zero MSE and zero risk on six high-utilization cells. Four pass; day-61 and day-114 contracts at nominal utilization 1.10 are infeasible. Minimum-MSE solves under unchanged risk and capacity certify achievable errors of 1.98×10^{-7} and 1.00×10^{-6} MW^2 respectively, with zero gap in all six cells. These retained failures show why runtime feasibility, risk, and accuracy must be enforced together rather than inherited from an aggregate fit.

C. Signed network value and security

Signed netting changes settlement more than substituting exact value for its linear approximation in the base case. For the trace-anchored rows in Table II, exact net-value MAE is \$151.719/day versus \$360.580/day for nodal gross response. The paired netting improvement is \$208.190/day (Holm-adjusted $p = 3.89 \times 10^{-4}$); the linear-to-exact change is \$0.671/day ($p = 0.803$). All 5,616 interval/counterfactual instances satisfy (32). These comparisons include the nodal-difference space–time structure of virtual-link remuneration,

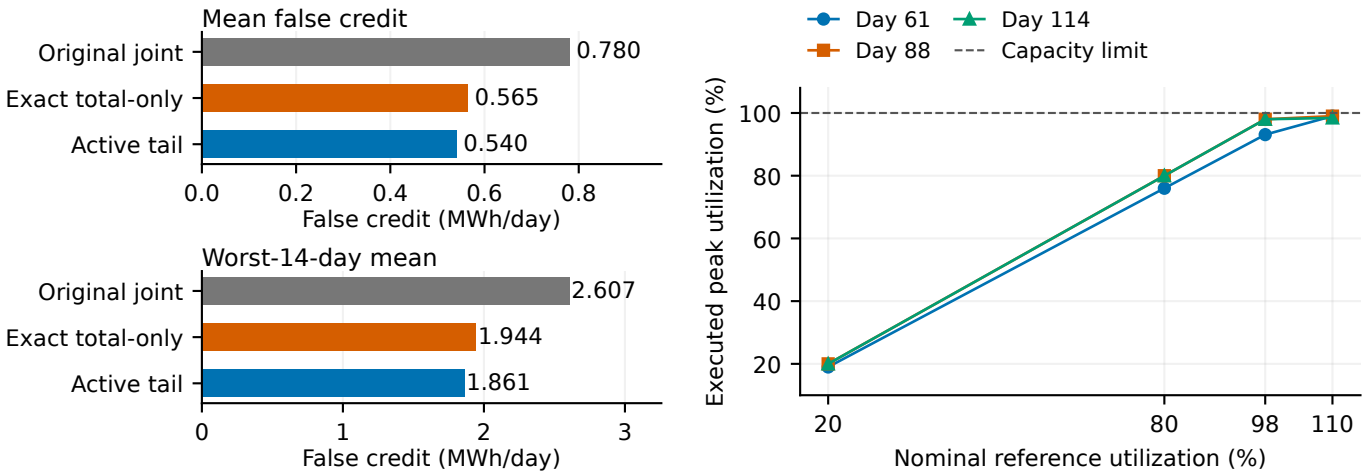


Fig. 3. Risk control and coupled execution. Left: original joint and exact-MSE controls on the retrospective 54-day panel; the active-tail versus exact-total-only tail difference has an interval including zero. Right: full-runtime, direct-risk schedules across the predeclared capacity scan. Each plotted point is globally optimal for its finite model; no full-population congestion claim is inferred from these 64-job cohorts.

TABLE II
N–1 SETTLEMENT ERROR ON A COMMON PHYSICAL RESPONSE (USD/DAY). OVERPAYMENT IS THE MEAN POSITIVE ERROR; MEDIANS ARE ABSOLUTE ERRORS.

		MAE		Median		Overpayment	
Panel	Settlement rule	Trace	Workload	Trace	Workload	Trace	Workload
N–1	Base-case exact	139.221	167.367	79.790	121.010	130.420	107.690
N–1	N–1 linear	2.208	150.461	0.600	113.430	1.910	20.380
N–1	N–1 exact	0.990	150.955	0.870	113.430	0.010	20.180

while the exact benchmark captures changes in dispatch regimes. For the workload response, base-case MAE is \$455.865/day for exact signed value and \$421.677/day for nodal gross response, so the trace-based gain is not universal.

The DC replay imposes all 37 finite non-islanding RTS-24 outages and reaches maximum loading 1.000 p.u. Its denominator is not filtered using method outcomes. The value-component cap passes all five frozen conversion scenarios in 54 daily programs within 3×10^{-11} USD. It does not dominate all comparators: at the median/high conversion factors, the value-certified verifier's MAEs are \$57.357/\$322.032 per day, versus the single projection's \$41.141/\$231.841. On two held-out interior factors the joint-risk verifier also has lower error than the value-certified profile. Contractual dominance of a specified value cap and predictive accuracy are different properties.

The AC audit retains a common topology-connected domain of 10,332 cells: 37/38/35/177 outages for RTS-24/IEEE-30/IEEE-39/IEEE-118, three penetrations, two methods, and six day–slot snapshots. It records 9,642 certified cells, 690 numerical unknowns, and no converged limit violations. The latter is not a proof that unknown cells are feasible. The revision reuses 9,576 checksum-bound certificates and solves 756 cells, including matched controls. On IEEE-30 outages 29 and 31, the proposed profile certifies all 36 cells; the reference certifies 30, with six unknowns at 9% penetration. Those six outcomes no longer remove the reference's 30 successful cells. Native screening is reported only as a subgroup; all methods retain the same primary denominator. Six snapshots test the stated operating conditions, not an annual reliability rate.

The original complete-cycle runtime replay reports signed value \$184.664 and positive cash presentation \$283.470 across 5,184 cells. Keeping these ledgers separate exposes negative network-value intervals that a cash floor would hide. They are declaration-only values, not payments: the lineage audit requires an independently closed meter before activation.

The old slot-62 gate and complete-ledger panels have different eligible populations and scoring tasks. Their F_1 values of 0.080 and 0.536 cannot be read as a matched causal degradation. The new matched protocol tests that boundary directly. The existing 79-job binding-capacity experiment uses calibrated service with a possible final partial slot; it does not establish congestion performance for the full-runtime nameplate model. Neither public trace replay nor synthetic geography can replace an identifiable utility event, authenticated declarations, and a prospective deployment trial.

VI. CONCLUSION

This paper develops a contract-based connection between data-center workload verification, runtime scheduling, and security-aware net-value settlement. The exact convex fit makes tail-risk control identifiable, while residual bounds and direct integer constraints establish when execution preserves a specified contract. The coupled solver maintains job power, duration, and deadlines and computes minimum achievable error when exact matching is infeasible. Signed accounting and a common AC contingency denominator make network-value and certification results comparable across methods. The

retrospective tests support a controllable risk–accuracy frontier and exact finite-scenario feasibility checks. Prospective utility-event validation and congested full-population testing remain the next steps for deployment.

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